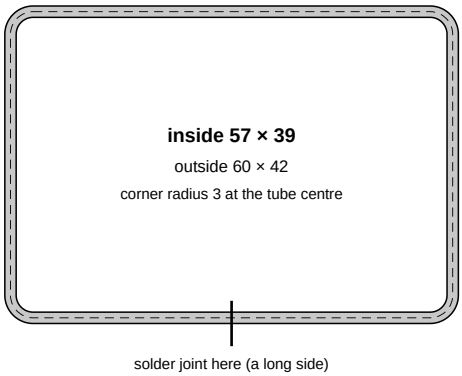
Pocket AI brass frame · stacked layout · 1:1 template (Letter)

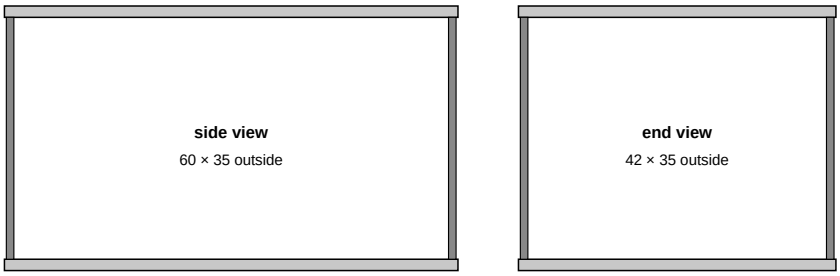
Print at 100 % / Actual size, Letter. Check the 100 mm ruler at the bottom before you bend or cut anything.

1 · LOOP · make 2 (top, bottom), 1.5 mm tube



Grey band = the tube. Dashed = tube centreline. Bend the tube to sit on the grey band.

2 · ASSEMBLED, 1:1

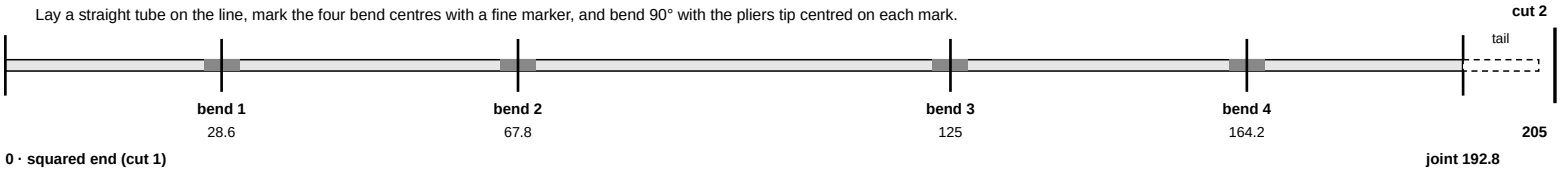


Inside height 32 between the loops = the corner posts.

Screen faces up through the top loop; A1 sits on the bottom layer.

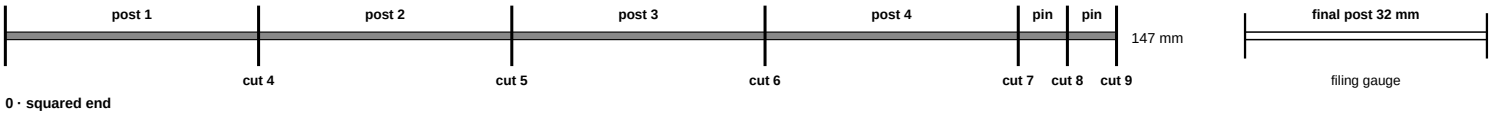
3 · TUBE MARKING STRIP · hollow 1.5 mm tube · one loop = 192.8 mm along its centre

Lay a straight tube on the line, mark the four bend centres with a fine marker, and bend 90° with the pliers tip centred on each mark.



Squared end on 0. Cut 2 at 205 before marking. Leave the dashed 10 mm tail on; cut it (cut 3) only after the four bends.

4 · ROD MARKING STRIP · solid 1.0 mm rod · 4 posts + 2 pins

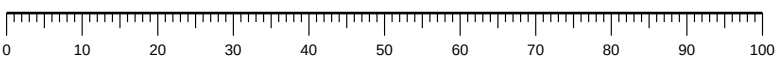


Squared end on 0; mark all six lines; saw on each. The extra 0.5 mm per piece is the saw cut. File each post to 32 mm at its own corner.

5 · CUT LIST

1.5 mm tube (K&S #9831)	2 loops × 202.8 mm with tail	one 300 mm tube per loop; 2 of your 4 tubes
1.0 mm rod (K&S #9861)	4 posts × 33.5 + 2 pins × 6.5 mm	147 mm of one 300 mm rod

6 · RULES

- Size: cardboard Box B inside 55 × 37 × 30 mm + 1 mm each side = 57 × 39 × 32 inside the brass.
- Test-bend an offcut first. If the thin tube kinks, bend on the thicker part of the round-nose pliers tip (near the hinge).
- Cut with the jeweler's saw on the bench pin, never flush cutters. Deburr every cut end with 400–800 sandpaper. Safety glasses on.
- Solder the EMPTY frame only (Harris acid flux). Then hot water, baking-soda rinse and 91 % alcohol before any electronics come near.
- The frame carries no current: it never touches ground, USB, either speaker wire or a bare pad. Line contact points with Kapton.
- Keep both loop joints on long sides, and keep A1's antenna end as far from brass as your layout allows. Check Wi-Fi after mounting.

SCALE CHECK: this ruler must measure exactly 100 mm. If not, reprint at 100 % / Actual size.